

Under-water ignition and combustion of PDMS processed Al/CuO nanoparticle composite

Z. B. Wang · A. Q. Wang · J. Ehling  · C. J. MacRobbie · S. Kim · L. Saini · S. Deng  · J. Z. Wen  · M. Y. Shen

Received: 15 April 2026 / Accepted: 24 August 2026

© The Author(s) 2026

Abstract

This study investigates the fabrication and performance of hydrophobic nanothermite powders prepared by combining liquid polydimethylsiloxane (PDMS) prepolymers with thermite powders in a dispersion medium. The resulting composite remained dry when submerged underwater and could be ignited by laser irradiation. High-speed imaging measurements showed an ignition delay of 2.3 ms under underwater conditions. Finite-element simulations based on a two-phase flow model revealed rapid cooling and condensation of water vapor during ignition. Preliminary analysis of the bubble composition suggests the evolution of gaseous products from the thermite reactions. These results demonstrate the feasibility of underwater laser ignition of thermite–PDMS composites and provide insights into the complex physical and chemical processes involved in underwater combustion.

Keywords Thermite reactions · Hydrophobic nanothermite · PDMS · Contact angle · Underwater combustion · Heat transport · Laser-induced ignition

Introduction

Thermite formulations represent a category of highly energetic materials with characteristic exothermic reactions, generating substantial heat upon ignition. Nanothermite refers to a solid mixture of nano-sized reactive particles. These compositions are commonly referred to as metastable intermolecular composites, denoting their distinctive reactive properties¹. The resulting high temperatures, combined with their ease of use and portability, have earned thermite great interest across various applications, including propulsion systems, controlled demolitions, welding and metal cutting, and firework displays. An easy way to initiate and sustain underwater combustion could offer significant potential for various marine and aquatic applications, including welding^{2,3}, cutting, and propulsion. However, the hydrophilic nature of aluminum⁴ and copper oxide⁵ causes wetting when submerged in water, leading to heat loss that hinders ignition and flame propagation. Consequently, research in underwater combustion is limited, with a lack of understanding of reaction propagation and energy release from nanothermite reactions underwater.

This Article in Press is shared early to give you faster access to new research. It is citable and carries a permanent DOI. The final edited version will replace it automatically.



They are most commonly composed of metal powder as a reducing agent and an oxidizer such as metal oxides, while fluoropolymers⁶ or iodine oxides⁷ can also be used. In this study, aluminum and copper oxide are used, and the reaction can be described as:



The initiation of thermite combustion typically requires a high-temperature heat source, such as spark or flame. Once activated, the reaction between the metal powder and metal oxide can reach temperature up to 2843 K for Al/CuO composites^{8,9}. This characteristic imparts thermite with remarkable effectiveness as a potent heat source.

The aluminum thermite system has been well studied, revealing two main modes of aluminum oxidation: one occurring in the condensed phase, where oxygen ions traverse the Al/CuO interface, and the other in the gas phase, involving the dissociation of CuO before O₂ gas engages with aluminum, resembling the oxidation process in an oxygen-rich environment¹⁰. Upon ignition of the Al/CuO mixture, the flame propagates through mass and heat transfer processes such as thermal conduction and mass convection. Experimental observations using electrically heated nanothermite coatings have indicated that CuO decomposition precedes ignition when the heating rate exceeds 2,000 K/s¹¹.

Additionally, lightly packed Al/CuO powders were observed to have three modes of combustion with increasing surrounding inert gas pressure, with combustion velocity decreasing from 1,000 to 2 m/s¹². Convection dominates at low pressures, and thermal conduction dominants at high pressures, suggesting a gas-solid reaction mechanism¹³.

In addition to these mechanisms, when the particle sizes drop to the nanoscale, nanothermites behave slightly differently from traditional ones, as the thickness of the oxidation layer on the aluminum powder can no longer be neglected, forming a shell of alumina, which not only reduces the effective aluminum weight fraction, but also inhibits the reaction. At this scale, Al/CuO nanothermite can also react through reactive sintering¹⁴, where reaction initiation occurs at the Al/CuO interface, and capillary forces sustain the process by delivering unreacted material to the reaction site. Reactive sintering occurs much faster than the burning time, typically within 10 μs.

Nevertheless, efforts on underwater combustion have been made. Meir et al. initiated combustion in a naturally hydrophobic iron thermite using microwaves. The sample was confined underwater within a bubble by a magnet using the bubble-marble effect¹⁵. Yu et al. employed a different approach by igniting superhydrophobic Co₃O₄/Al core-shell nano-wires with a nanosecond Nd/YAG laser. The fabrication involved vacuum-coating aluminum onto Co₃O₄ nanowires, before applying a hydrophobic fluoroalkylsilane layer. While these pioneering methods show promise, they exhibit limitations, the iron thermite is not easily generalized to other components, and the nanowires involve a complicated fabrication that is slow and expensive and are limited to two-dimensional structures.

This paper presents a simple method to achieve underwater combustion of thermites by introducing a novel approach for crafting hydrophobic nanothermite powders. The sample is made by mixing liquid PDMS prepolymers with thermite powders within a dispersion liquid. The resultant composite remained dry when submerged underwater and could be readily ignited by laser irradiation. Furthermore, the concept of creating a hydrophobic coating using pre-curing polymers could be applied to materials beyond CuO/Al thermite or PDMS.

Beyond demonstrating underwater ignition and combustion, this research investigates the mechanisms in Al/CuO nanothermite when immersed underwater with PDMS as an additional reagent, to deepen the understanding of heat dissipation channels, water phase change with rapid heating, and potential gas generation.

Sample preparation and characterization

To prepare a thermite sample with an equivalence ratio of 1 (ER = 1), 0.8 g CuO and 0.27 g Al nano-powder (~50 nm diameter, SkySpring Nanomaterials, Inc.) were used. The excess aluminum compensates for the oxidation layer on the Al nano-powders. The prepolymer and catalyst (Dow Corning Sylgard 184 kit) were mixed thoroughly at 10:1 mass ratio. The powders and about 60 mg of the prepolymer were added to a beaker followed by the addition of approximately 5 ml acetone to fully immerse the powders. The mixture was ultrasonicated and thoroughly mixed then dried and cured for 2 days in a wide container under well-ventilated conditions. The sample dried and naturally cracked into ~3 mm pieces.

The SEM images were taken by a JEOL JSM 7401 F unit and the results are shown in Fig. 1. The raw thermite sample is presented in Fig. 1a, while the 5% PDMS composite is shown in Fig. 1b. Compared to the raw thermite sample, the morphology does not change significantly, although the size of the particles may have increased slightly. If the PDMS is assumed to be evenly coated on each particle, the thickness is estimated to be less than 2 nm for a 5% PDMS mass ratio.

By submerging the sample with 5% PDMS ratio in 1% hydrochloric (HCl) acid for several days, the acid slowly dissolves the CuO and Al particles, leaving behind the PDMS framework that binds these nanoparticles, and its SEM image is shown in Fig. 1f. The spherical cavities visible in the green box in Fig. 1f suggest that PDMS acts as a binding agent, clustering the nanoparticles into distinct clumps rather than uniformly coating each individual particle. This indicates that the CuO/Al nanoparticles are still in contact with each other, and existing combustion mechanism, such as reactive sintering may still be valid with the introduction of the PDMS coating. The smaller bright spots within the green box in Fig. 1f are sputter-coated gold particles for better imaging.

The coated sample exhibits visual similarity to the original sample, and the evaluation of its hydrophobicity was conducted through contact angle measurements¹⁶. Upon gently pressing the powder into a 5-mm flake, the contact angle is measured to be approximately 150° for an 11.7% PDMS mass ratio, as illustrated in Fig. 2. This value is greater than the typical contact angle of PDMS, which is around 110°¹⁷. Sabbah et al. have previously reported an increase in the contact angle for PDMS with wrinkles, attributed to surface roughness¹⁸. The observed superhydrophobic effect in our study can be explained by the surface roughness induced by the thermite nanoparticles. Li et al. have also documented the fabrication of a superhydrophobic surface by coating PDMS on aluminum, aligning with the observed hydrophobic behavior in our coated thermite sample¹⁹.

The reported values correspond to static contact angles measured under ambient conditions. Dynamic contact angles (advancing/receding) and hysteresis were not measured, but are expected to influence wetting transitions and long-term stability of the hydrophobic state, as discussed by Butt et al.²⁰. These effects will be investigated in future work.

The optical reflection spectra (250–2500 nm) of Al nanoparticles, CuO nanoparticles, and their composite mixtures were measured using a UV–Vis–NIR spectrophotometer equipped with an integrating sphere attachment, designed specifically for rough pellet surfaces. Kramers–Kronig (KK) analysis typically assumes specular reflectance at a well-defined interface, such as the surface of a bulk solid. Nevertheless, it has been successfully applied to porous and nanostructured films, where the surfaces are rough, but the reflectance remains predominantly specular. Using KK analysis, we obtained the corresponding extinction spectra and found that the extinction coefficient of the Al–CuO nanocomposite is lower than the value predicted by a simple linear combination of the individual components. Furthermore, the characteristic absorption bands in the spectra shift to lower energies relative to those of the individual components [support data]. These spectral changes suggest that the close contact between Al and CuO nanoparticles influences the optical response of the composite. However, the optical measurements alone do not provide direct evidence of electronic coupling, charge transfer, or carrier-mediated interactions.

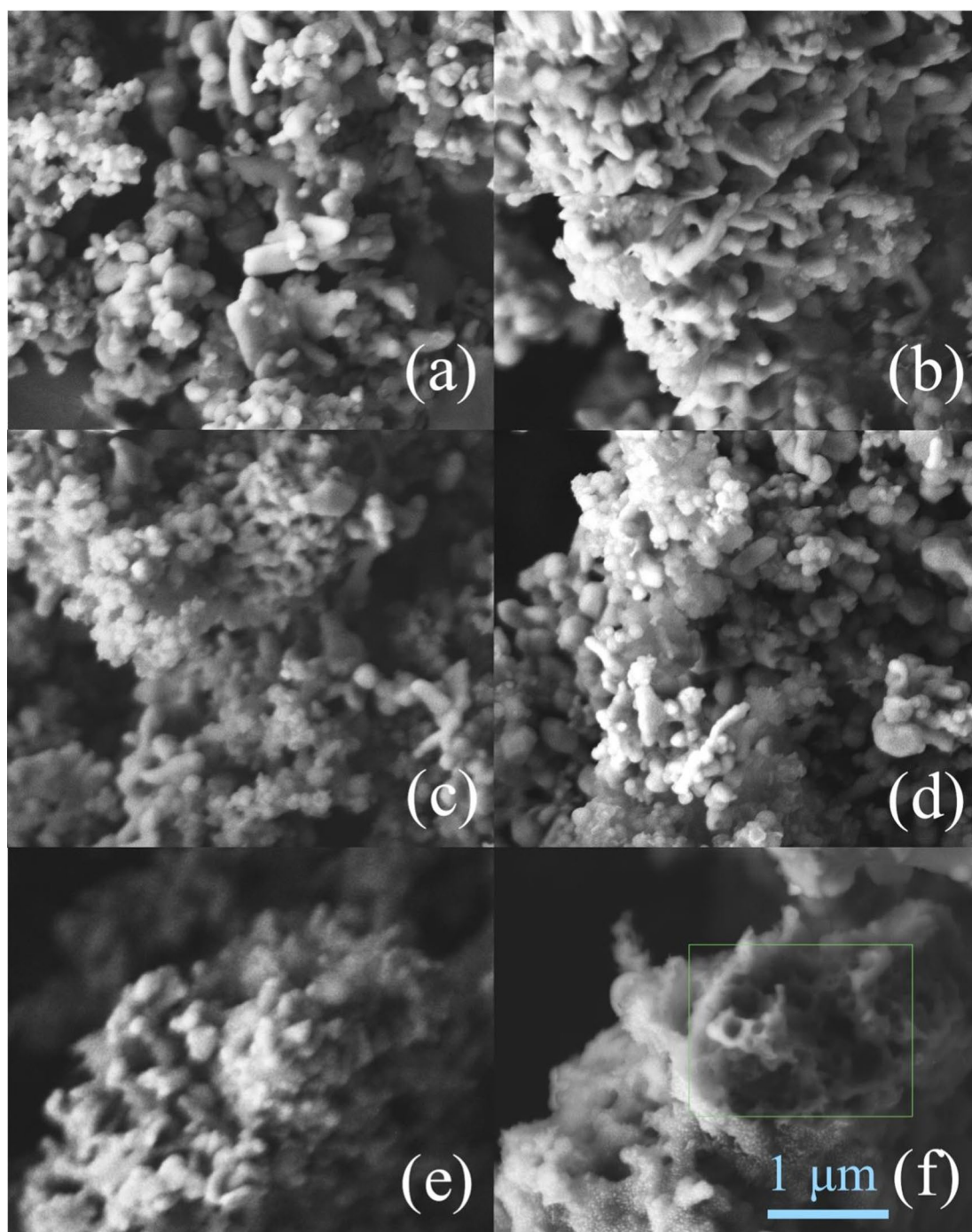


Fig. 1 SEM images of coated thermite with (a) 0%, (b) 2%, (c) 5%, (d) 8% and (e) 11.7% PDMS. After dissolving the Al/CuO-PDMS composite in a dilute HCl acid, the SEM image of the remaining PDMS is shown in (f). The spherical cavities in the green box indicate that the PDMS filled in the gaps of the particles. All the figures are on the same scale.

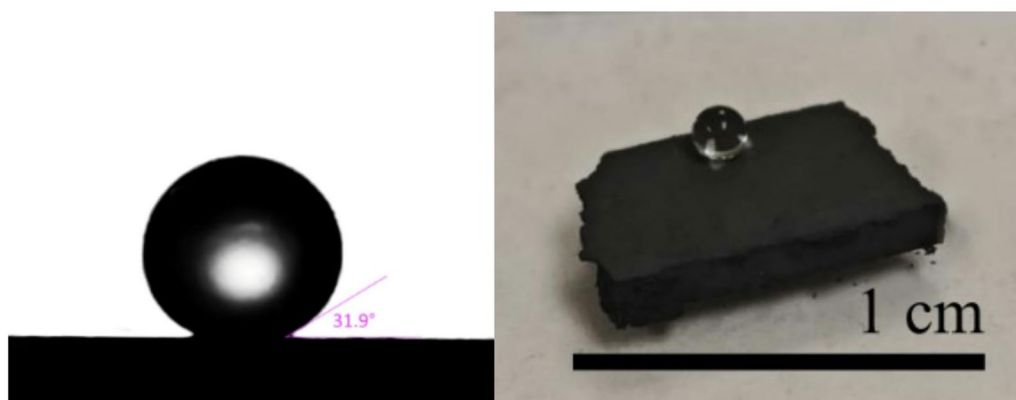
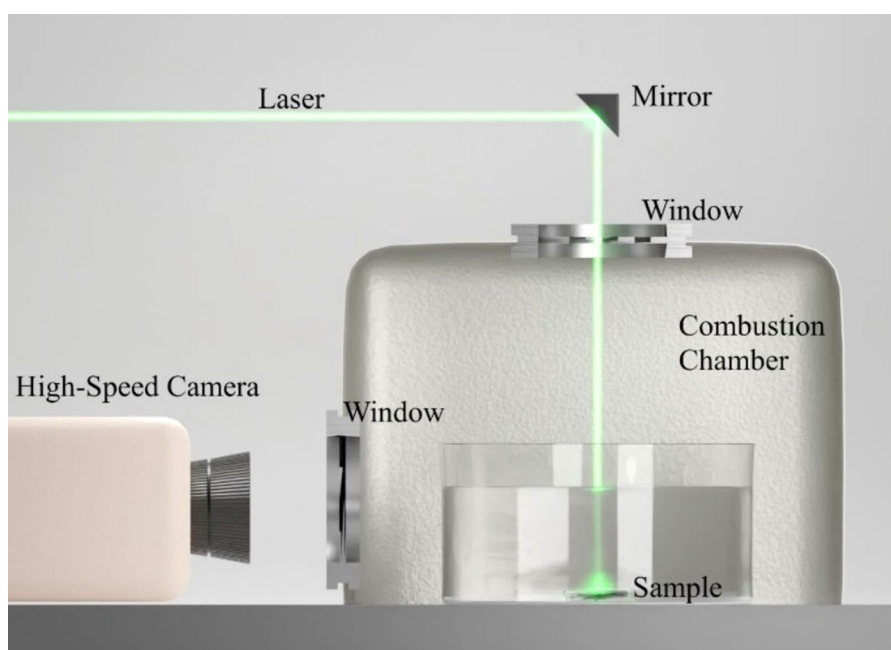


Fig. 2 The PDMS coating can make the thermite super-hydrophobic. Thermite with 11.7% PDMS ratio has a contact angle of $148.5^{\circ} \pm 4.7^{\circ}$.

Fig. 3 The hydrophobic thermite flake was placed in a container and water was added slowly. The laser and the camera were triggered simultaneously when the reaction started in the combustion chamber.



Experiments and results

Experimental setup and ignition conditions

As shown in Fig. 3, a piece of the flake (about 3 mg) was placed into a container before water was added. The sample was then placed in a combustion chamber and a 5 W continuous laser with 532 nm wavelength was used to ignite the sample. The laser spot diameter was approximately 200 μm , corresponding to a power density of approximately $1.6 \times 10^4 \text{ W/cm}^2$. The 532 nm wavelength was chosen due to the strong absorption by CuO and the Al/CuO composite, enabling efficient photothermal heating and rapid ignition. A function generator triggered the laser and the camera simultaneously.

To avoid ambiguity, the definitions of ignition and combustion used in this study are clarified as follows. Ignition is defined as the onset of a self-sustained exothermic reaction, identified by a sudden increase in emission intensity together with rapid gas generation. Combustion refers to the subsequent sustained reaction phase

involving continued heat release and product formation. It should be noted that bubble formation alone is not considered sufficient evidence of ignition but is instead interpreted in conjunction with optical/emission signatures observed in high-speed imaging.

Ignition behavior and delay time

Selected frames from the high-speed video are shown in Fig. 4. The complete video is provided in the Supporting Information. Figure 4b, c show that ignition occurred about 2.2 ms after the laser onset, delivering 11 mJ of energy to the sample. This is comparable to another experiment using a 1040 nm, 100-fs pulsed laser with a 10-MHz repetition rate and 4 W average power, which also ignited the sample in 2 ms. At a lower equivalence ratio (ER = 0.6), insufficient fuel significantly delayed the ignition to around 30 ms for both the green laser and the infrared laser. This trend is consistent with previous studies on Al/CuO nanothermites, where fuel-lean mixtures exhibit longer ignition delays compared to fuel-rich compositions due to reduced heat release and slower reaction kinetics. Additionally, ignition delay is known to strongly depend on equivalence ratio, with lean mixtures generally increasing ignition delay because of reduced reactive species concentration and slower chain-branching processes¹.

Bubble growth and dynamic evolution

To better analyze the reaction, Fig. 5 illustrates the temporal evolution of the bubble's size, where the horizontal size is indicated by the dashed arrow in Fig. 4g, and the vertical size is indicated by the solid arrow in Fig. 4h.

The bubble first appeared at 2.3 ms, as shown in Fig. 4c, initiating a slow-expansion stage at a speed of about 0.29 m/s, forming a 5-mm bubble by 5.3 ms, as shown in Fig. 4e. The bubble then expanded rapidly and reached a

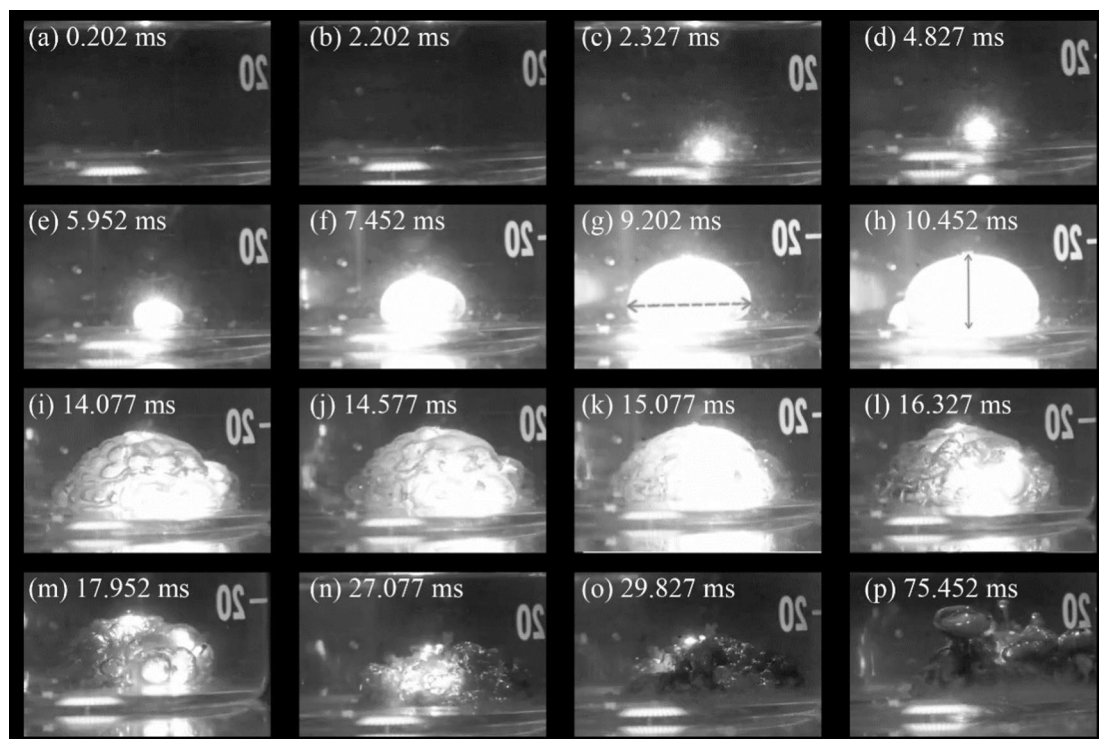
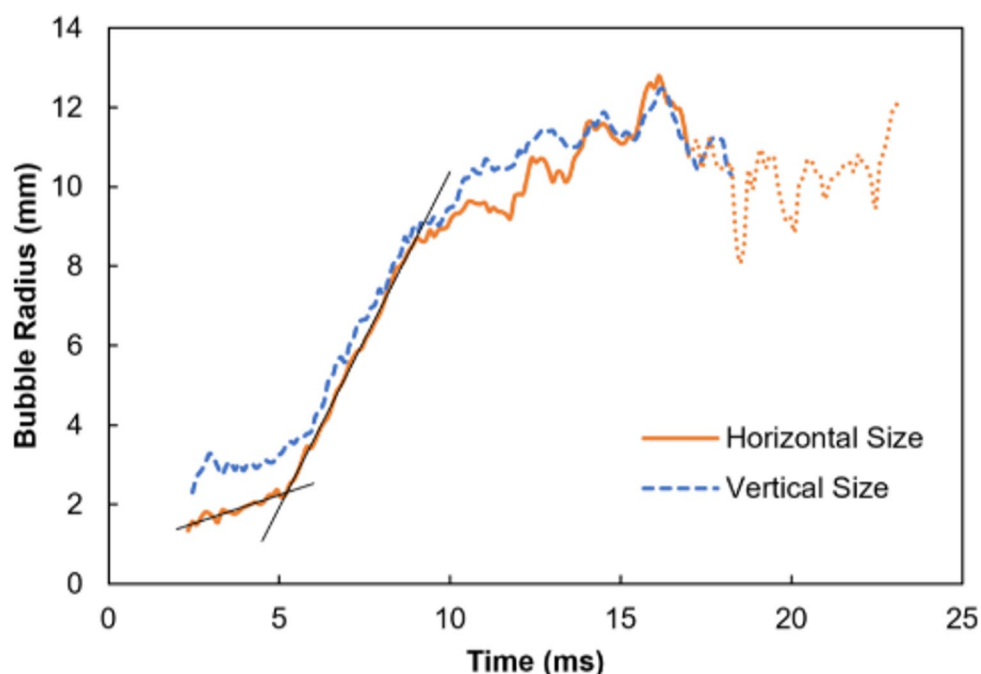


Fig. 4 High-speed images of the underwater ignition. Time in milliseconds after the laser onset is labeled. The horizontal size is indicated by the dashed arrow, and the vertical size is indicated by the solid arrow.

Fig. 5 Bubble size as a function of time. The vertical and horizontal direction give similar results, and the bubble expands before entering a pulsing stage after 10 ms, before completely collapses after 17 ms (dotted line).



diameter of 20 mm at a rate of 1.54 m/s. As discussed later, this slow expansion stage may be attributed to PDMS decomposition and vaporization while the fast expansion stage may be driven by the thermite reaction.

Following this fast-expanding phase, the bubble entered a pulsing stage characterized by size oscillations with a decelerating expansion, followed by fragmentation into smaller bubbles at the surface before completely collapsing. A more detailed discussion of this bubble generation process will be provided in the later discussion.

After the expansion stage, the bubble entered a pulsation stage where its volume oscillates around a baseline, as shown in Fig. 5 from 10 ms to about 25 ms. The oscillation frequency was determined to be approximately 600 Hz, which will be discussed later.

Small bubbles observed in Fig. 4 are attributed to rapid gas nucleation and fragmentation processes occurring during ignition and subsequent reactions. These bubbles likely originate from localized gas release and the breakup of the primary bubble during its dynamic evolution. Due to their transient nature, small size, and limited spatial resolution in the high-speed imaging, these secondary bubbles were not included in the quantitative analysis. The analysis presented here therefore focuses on the primary bubble, which captures the dominant features of the bubble growth and oscillation dynamics.

The results presented in Figs. 4 and 5 correspond to a representative experiment for a PDMS mass ratio of approximately 12%. While additional ignition experiments confirmed similar qualitative behavior, the detailed quantitative bubble analysis was performed on a single high-speed recording obtained from a small (~3 mg) sample. Larger-pellet experiments produced substantially more gas and more complex bubble structures, making quantitative tracking difficult. A systematic investigation of repeatability, uncertainty, and the influence of sample mass and PDMS content will be the subject of future work.

Effect of hydrophobicity and aging

Breaking the flakes into powder or pressing them into pellets does not prevent the sample from being ignited. However, the flammability is greatly inhibited when water permeates into the sample, either by degassing or natural aging. The aging test involved submerging the coated thermite samples underwater for 2 days, and all the samples lost their ability to be ignited or to sustain a flame underwater. As Samaha et al. concluded²¹, superhydrophobic

surfaces usually contain micro- or nanostructures within which small air pockets can be trapped. In this case, this entrapped air could dissolve into water when the surface is submerged, causing the surface to lose its hydrophobicity over time. In this study, this aging effect also suggests that the ability to be ignited underwater is a result of superhydrophobicity, and isolation from water is critical by reducing the contact and the heat drain from the water environment, so that it behaves more similarly to conventional conditions, where the sample could be ignited easily in air and has been extensively studied.

This aging effect is also expected to be more significant at higher pressures, as pressure increases gas solubility²¹. The loss of ignition after aging suggests gradual water infiltration and depletion of trapped air pockets, which indicates limited shelf life under submerged conditions. Potential strategies to improve stability include thicker coatings, increased crosslinking density, or alternative encapsulation approaches.

Effect of PDMS content on combustion behavior

Samples with different PDMS ratios were also tested, and their respective contact angles were measured, as shown in Fig. 6. Each data point represents the average of about 6 measurements, with error bars indicating the standard deviation. The contact angle demonstrates a notable upward trend with increasing PDMS coating thickness. While most samples remain ignitable by laser, a transition occurs around a mass ratio of ~12%, beyond which the sample become fire-retardant. The SEM images in Fig. 1d suggest increased clumping, attributed to excess PDMS, potentially explaining the observed flame-retardant properties. At a ~1% PDMS ratio, rapid water permeation occurs, preventing ignition by lasers irradiation (blue triangle). As the PDMS content increases, the reaction tends to be less vigorous, with a higher likelihood of incomplete combustion at elevated PDMS ratios (green circles). The SEM images in Fig. 1a-d suggest that PDMS content does not significantly alter the morphology of the sample. These combustion characteristics suggest that PDMS retains flame-retardant properties, and an excess amount inhibits the reaction. Since the nanostructures of the samples remain similar, the thickness of the PDMS coating might be the key factor causing these differences, which can be investigated in future work.

Spathi et al. dry-milled paper sludge ash with stearic acid and reported similar contact angles and a comparable increasing trend with mass ratio²², which matches our observations. This suggests that the increase in contact angle when powders are coated with hydrophobic substances is a general phenomenon rather than a distinctive

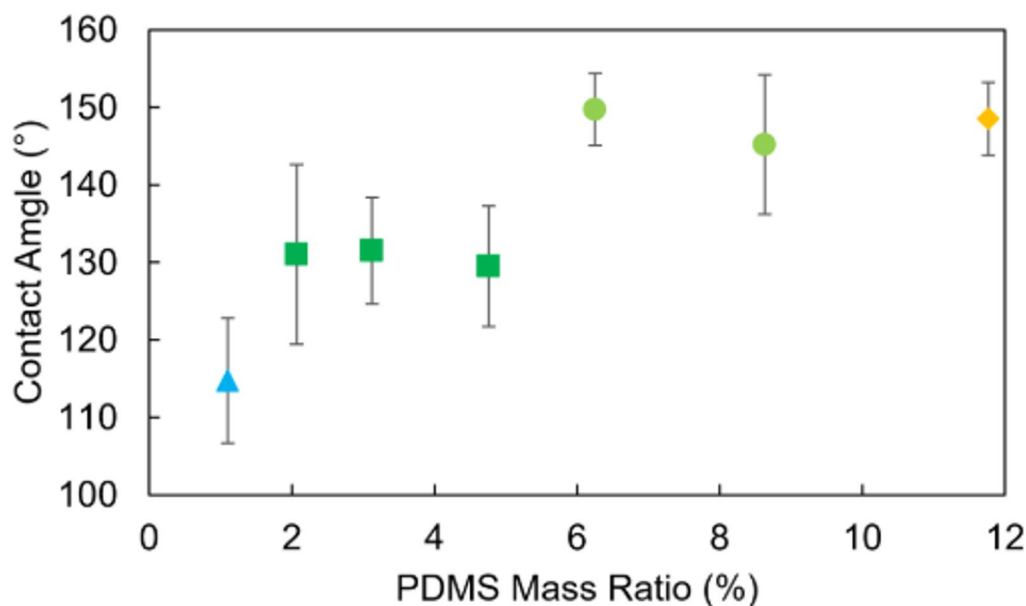


Fig. 6 Contact angle of different PDMS ratio, a rough increasing trend could be observed.

feature of PDMS. This opens up possibilities for enhancing the performance of our samples and methods by exploring other hydrophobic and combustible substances.

In addition, substituting aluminum nanopowder with micro-sized particles does not yield the same underwater burning effect under identical conditions. This observation suggests that the nanoscale characteristics of the Al particles may play an important role in enabling underwater ignition under the present experimental conditions. The underlying mechanisms, including possible nanoscale interfacial and reaction processes such as reactive sintering, remain to be established through further investigation.

The results presented in Figs. 4 and 5 correspond to representative experiments for a PDMS mass ratio of approximately 12%. While consistent qualitative behavior is observed, parameters such as ignition delay, bubble size, and oscillation dynamics vary with composition and experimental conditions. A systematic study of repeatability, uncertainty, and the influence of PDMS ratio is ongoing and will be necessary to provide a comprehensive statistical characterization of the system.

Analysis and discussion

Role of PDMS in underwater ignition

The role of PDMS in the present system is multifaceted. Experimentally, its primary function is to provide a hydrophobic coating that facilitates underwater ignition by reducing direct contact between water and the thermite particles. However, additional effects may also contribute, including thermal insulation, interfacial modification, and possible gas generation during decomposition. The relative importance of these mechanisms cannot be resolved based on the current data, and their contributions are coupled under the transient high-temperature conditions of ignition. A more detailed understanding of the role of the polymer coating would require systematic studies, such as comparisons with non-reactive hydrophobic coatings, systematic variations in coating thickness and composition, and direct characterization of thermal and gas-generation behavior. These investigations are beyond the scope of the present work and will be addressed in future studies.

Bubble dynamics modeling

During the nanothermite reaction, the initial nanoparticles are rapidly transformed into Al_2O_3 and Cu products, which are generated in a highly transient and expanding environment. A significant portion of these particles is entrained within the bubble or concentrated near its interface rather than uniformly dispersed in the surrounding liquid. While studies have shown that clustered nanoparticles can significantly modify local rheological behavior even at low concentrations^{23,24}, in the present system such effects are expected to be localized. Consequently, the bulk liquid may still be approximated as Newtonian to first order, while localized particle–fluid interactions near the interface may enhance viscous dissipation, damping, and deviations from idealized bubble dynamics. These effects are not explicitly included in the current model and represent a source of uncertainty.

The effect of ambient pressure was not explicitly investigated in this work. At increased water depth, higher pressure is expected to enhance gas solubility and suppress bubble formation and persistence. This may reduce the effectiveness of the hydrophobic air layer and increase heat loss to the surrounding liquid, potentially inhibiting ignition and sustained combustion. Therefore, the present approach may have limited applicability under high-pressure underwater conditions.

The Rayleigh–Plesset equation describes the radial dynamics of a gas bubble in a liquid by balancing inertial, viscous, surface tension, and pressure forces acting on a spherical interface. It is derived under the assumptions of a perfectly spherical bubble in an infinite, incompressible liquid, with symmetric radial motion. In the present study, the observed bubble is not perfectly spherical but rather approximately hemispherical and attached to a solid surface, which introduces geometric asymmetry and boundary effects. These conditions violate the strict

assumptions of the Rayleigh–Plesset formulation and can modify the effective inertia, damping, and pressure distribution around the bubble. In particular, the presence of the solid boundary reduces the amount of liquid that must be displaced during expansion and can lead to higher oscillation frequencies and altered collapse dynamics.

Despite these limitations, the Rayleigh–Plesset equation is employed here as a first-order, reduced-order model to capture the dominant features of bubble growth and oscillation. Previous studies^{25,26} have shown that, even under non-ideal conditions, it can provide reasonable order-of-magnitude estimates of bubble dynamics when interpreted qualitatively. Accordingly, any gas temperatures, pressures, or gas quantities inferred from the model should be interpreted as approximate estimates subject to the assumptions of the analysis rather than directly measured quantities.

For a more thorough examination of the evolution of the bubble, the Rayleigh–Plesset equation can be employed:

$$R \frac{d^2 R}{dt^2} + \frac{3}{2} \left(\frac{dR}{dt} \right)^2 + \frac{4\nu_L}{R} \frac{dR}{dt} + \frac{2\sigma}{\rho_L R} + \frac{\Delta P(t)}{\rho_L} = 0 \quad (2)$$

where ρ_L is the liquid density, $R(t)$ is the bubble radius, ν_L is the liquid kinematic viscosity, σ is the surface tension of the bubble interface, and $\Delta P(t) = P_b(t) - P_\infty$ is the pressure difference between the bubble interior and the far-field liquid.

Thermodynamic analysis and gas behavior

The evolution of the bubble is accompanied by continuous nanothermite reactions and heat exchange with the surrounding environment. To gain insight into this complex process, rather than attempting to isolate the contribution of each individual mechanism, the overall system behavior was analyzed in terms of thermodynamic parameters. The pressure within the bubble, P_b , can be calculated using the ideal gas law:

$$P_b = \frac{nk_B N_A T}{V} \quad (3)$$

Where n is the number of moles of the gases inside the bubble, k_B is the Boltzmann constant, N_A is the Avogadro constant, T is the temperature of the bubble as a function of time, and V is the volume of the bubble which can be derived from the radius R .

After setting values for the temperature and gas quantity inside the bubble, the radius of the bubble can be determined by solving the Rayleigh–Plesset equation.

To assess the plausibility of the assumed gas content, an order-of-magnitude energy-balance analysis was performed. For a typical sample mass of approximately 3 mg and an experimentally measured heat release of 1030 J/g obtained from the DSC measurements, the total released energy is approximately 3.1 J. If this energy were used entirely for water vaporization, the maximum amount of water vapor produced would be approximately 7.6×10^{-5} mol. This value is of the same order of magnitude as the gas quantity used in the bubble model ($\sim 10^{-5}$ mol), suggesting that the assumed gas content is physically reasonable. Since heat losses to the surrounding liquid, sensible heating of the products, and bubble-expansion work are neglected, this estimate represents an upper bound on the amount of vapor generated.

Based on this analysis and the experimentally observed bubble volume after 20 ms, a representative total gas quantity of 3.8×10^{-5} mol was assumed at 10 ms, comprising approximately 2.6×10^{-5} mol of water vapor and 1.2×10^{-5} mol of other gases. It should be emphasized that these gas quantities are not directly measured. Rather, they are representative values selected within physically reasonable bounds to reproduce the observed bubble dynamics through the Rayleigh–Plesset analysis and thermodynamic considerations. Under these assumptions, the calculated effective gas temperature can reach approximately 750 K. This temperature should be regarded as a model-derived effective value rather than an experimentally measured gas temperature.

The choice of temperature and gas quantity is not unique, as their product determines the bubble pressure through the ideal gas law. The maximum temperature is constrained by the nanothermite reaction temperature and

is therefore expected to remain below 2843 K for Al/CuO thermite combustion^{8,9}. The simulation matches the experimental results well before 10 ms, capturing both the overall growth profile and the small oscillations superimposed on the expansion between 6 and 11 ms. However, beyond 10 ms, although the overall bubble size can be reproduced, the predicted oscillation frequency deviates from the experimental observations when a slowly varying temperature or gas quantity is assumed. Consequently, the gas temperatures and gas quantities reported here are best interpreted as order-of-magnitude estimates that provide a physically plausible description of the bubble evolution rather than unique solutions or directly measured properties of the system.

Oscillation behavior and frequency analysis

To further characterize the oscillatory behavior of the bubble, the natural frequency can be estimated using Minnaert's formulation. It should be noted, however, that this model is derived under the assumptions of a spherical bubble undergoing small-amplitude oscillations in an infinite, incompressible, and Newtonian liquid. In contrast, the bubbles observed in the present study are millimeter-scale, approximately hemispherical, and attached to a solid boundary, and they undergo large-amplitude oscillations with continuous gas generation and heat transfer. These conditions deviate from the assumptions of the Minnaert model and can affect the effective inertia, damping, and restoring forces governing the oscillation.

Therefore, the Minnaert equation is employed here only as a first-order, order-of-magnitude estimate of the oscillation frequency. Deviations between predicted and observed frequencies are expected due to geometric asymmetry, boundary effects, and nonlinear oscillation dynamics. The natural bubble frequency could be estimated by a formula proposed by Minnaert in adiabatic conditions²⁷:

$$f = \frac{1}{2\pi R_0} \sqrt{\frac{3\gamma P_0}{\rho_L}} \quad (4)$$

Where γ is the specific heat ratio of the gas, R_0 is the steady-state radius of the bubble, P_0 is the steady-state pressure and ρ_L is the density of the liquid. If the bubbles are small and the process is assumed to be isothermal, the factor γ can be set to one which reduces the frequency by $\sim 16\%$ ²⁸.

Equation (4) yields an estimated frequency of approximately 360 Hz, which is consistent with the simulated result but is about half of the experimentally observed value. This discrepancy can be attributed to several factors. First, the bubble does not exist in an infinite liquid environment as assumed in Minnaert's derivation, and the presence of a nearby solid surface reduces the effective inertia of the surrounding liquid. In addition, the bubble is approximately hemispherical rather than spherical, resulting in asymmetric boundary conditions that are known to increase the oscillation frequency.

Furthermore, the bubble shape evolves over time and deviates from the spherical assumption, particularly after 14 ms during the collapse stage. The Minnaert model also assumes a uniform bubble with small-amplitude oscillations, whereas in the present system the bubble undergoes large-amplitude dynamics with continuous gas generation and heat exchange. These non-ideal and transient effects introduce additional complexity and contribute to the observed deviation between the predicted and measured frequencies.

Equation (4) implies that the natural frequency of the bubble only depends on the steady-state radius. Thus, under the assumption of slowly changing temperature and gas quantity, the bubble frequency could not be matched. However, it is possible to simulate the observed bubble evolution using the Rayleigh-Plesset equation. By smoothening the measured bubble radius and plugging it into the Rayleigh-Plesset equation, the pressure term could be obtained. The product nT after 10 ms can then be determined by the ideal gas law. Although both the temperature and gas quantity may vary with time, the same temperature evolution used in the earlier analysis was retained for simplicity, and the results are shown in Fig. 7b.

If the Rayleigh-Plesset equation is still assumed to be valid before the bubble completely collapses, the findings suggest that the bubble radius between 10 and 17 ms is a forced oscillation driven by the gas quantity. This

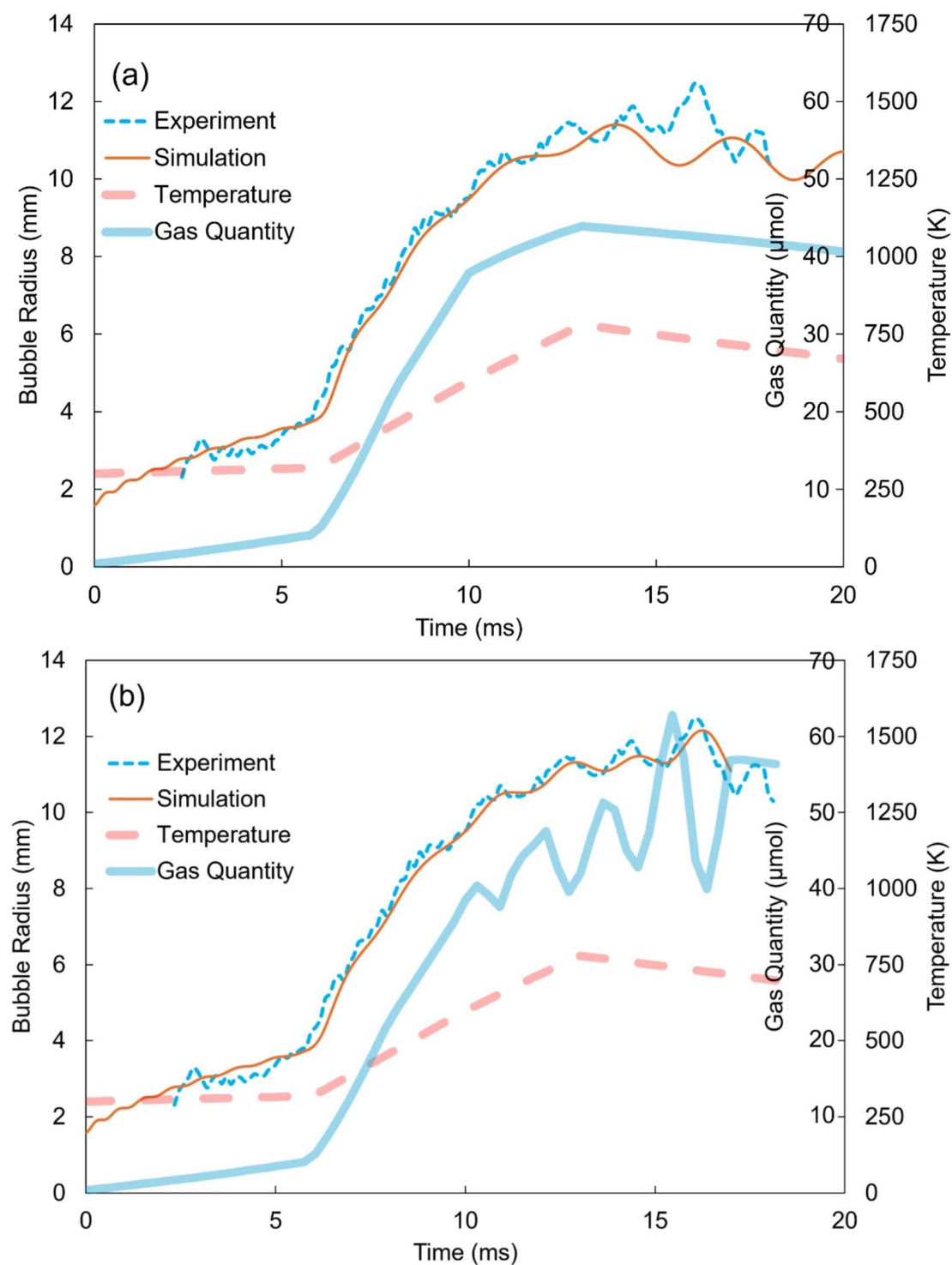


Fig. 7 Simulated bubble evolution based on the Rayleigh–Plesset model. **(a)** Calculated effective gas temperature and assumed gas quantity as a function of time. **(b)** Estimated gas quantity obtained by fitting the model to the measured bubble dynamics. The temperatures and gas quantities shown are model-derived estimates based on thermodynamic assumptions and bubble-dynamics analysis and do not represent direct experimental measurements.

oscillation in gas quantity could result from gas generated by PDMS decomposition, nanothermite reactions, and evaporation, and gas consumed by the condensation of PDMS oligomers and water vapor. However, this potential resonance needs stronger evidence and additional literature support. Consequently, the proposed forced-oscillation mechanism should be regarded as a plausible explanation rather than a definitive conclusion. The previous discussion of a shifted bubble frequency is considered the cause of such pulsations.

Gas generation mechanisms

Thermogravimetric analysis (TGA) and differential scanning calorimetry (DSC) can provide further insight into the behavior of solid combustion under water. TGA/DSC results were measured by a Simultaneous Thermal Analyzer (Netzsch STA 449 F1 Jupiter®). For each experiment, about 5 mg sample was placed in an alumina pan with lid, heated to 1500 °C at a rate of 20 °C/min. The samples were also protected by argon at a flow rate of 200 mL/min.

As shown in Fig. 8, the DSC of Al/CuO/PDMS shows the main exothermic reaction of nanothermite around 600 °C. The energy release is 1030 J/g, which is lower than the theoretical 4080 J/g⁹. This discrepancy is normal as the energy release measured by DSC is usually much lower than the theoretical and combustion values due to incomplete reaction¹¹. The relatively slow heating rate in DSC measurements differ significantly from rapid ignition conditions, which can alter reaction pathways and reduce the extent of reaction, thereby contributing to lower measured energy release¹⁰. Additionally, the incorporation of PDMS may further contribute to incomplete reaction, which requires more detailed investigation in future work.

A small endothermal dent was found at around 224–294 °C, which may be credited to the small amount of PDMS in the system that decomposed. The reaction temperature loosely matches the range of the PDMS decomposition. Therefore, under flash pyrolysis conditions, the nanothermite reaction temperature is high enough to decompose the PDMS. Besides, the small endothermal reaction at 1085 °C is the melting of Cu produced from the nanothermite reaction.

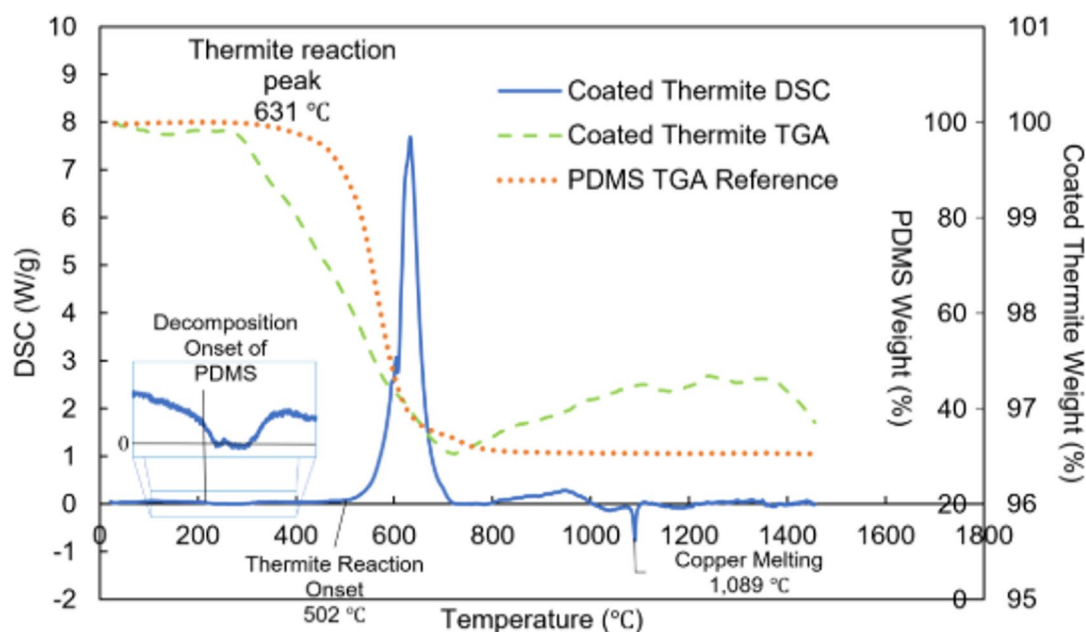


Fig. 8 DSC results (solid line) and TGA results (dashed line) of Al/CuO-PDMS composite (coated nanothermite). A dotted line also shows the TGA results of pure PDMS. The DSC result between 100 and 400 °C is magnified in the blue box to better show the small endothermal dent that may be PDMS decomposition.

The TGA results of the coated nanothermite and pure PDMS are shown in Fig. 8 in the dashed and dotted line. From the solid line, pure PDMS decomposes between 300 and 700 °C and loses about 70% of its weight²⁹. As Camino et al. reported, the decomposition of PDMS generates linear and cyclic oligomeric products from depolymerization of PDMS, which have gases generated from this decomposition process, which may be what remained at the end of the reaction, as shown in the high-speed images Fig. 4o, p, and will be discussed further.

The TGA of Al/CuO/PDMS showed a weight loss for the decomposition of PDMS. The decomposition was about 4%, close to the 70% value of 5% PDMS in the material. A minor weight gain was found as the reaction progress, which is due to the residue oxygen in the system that reacted with aluminum and caused weight gain. Although the system was vacuumed for multiple times before supplying Ar, there could still be some residue oxygen in the system, especially in the porous media such as alumina pans.

With this information, a preliminary analysis of the bubble's composition can be attempted. The generated gases in the final state illustrated in Fig. 4p is estimated to be approximately 10 small bubbles with a total volume of about 0.1 mL. On the other hand, the 0.25 mg PDMS in the sample, under standard conditions, is anticipated to generate approximately 0.01 mL of gas, assuming the gas products are oligomers with an average degree of polymerization of 4 in a fast-reacting scenario³⁰, resulting in a molar mass of approximately 300. However, the boiling points of these oligomers are around 140 to 290 °C³¹, which are significantly higher than the boiling point of water. According to the previous simulation, the bubble undergoes rapid cooling, suggesting that these oligomers would not remain in a gaseous state.

A different simulation by Bellama et al. suggests that smaller molecular species may be generated under rapid heating conditions³², and that PDMS decomposition under irradiation can produce light hydrocarbons such as methane or ethane through methyl radicals generated by Si–C bond cleavage^{32,33}. However, the extent to which such species are produced under the present conditions cannot be determined from the current experiments. Based on an idealized estimation, the maximum gas volume generated solely from PDMS decomposition would be on the order of ~ 0.1 mL, which is comparable to the observed bubble volume. This comparison suggests that additional gas sources may contribute; however, no direct measurement of gas composition was performed, and therefore this interpretation remains uncertain.

Other possible gas sources may include decomposition products associated with the nanothermite reaction. For example, partial decomposition of CuO could, in principle, release oxygen under certain conditions, although gas-phase oxygen generation is not required for nanothermite reactions³⁴. In addition, PDMS may undergo further reactions in the high-temperature environment, potentially forming gaseous products such as CO or CO₂³⁵.

In addition to gas generation from PDMS decomposition and possible contributions from the nanothermite reaction, the formation of water vapor due to rapid heating of the surrounding liquid is expected to play a significant role in the bubble dynamics. Given the substantial temperature difference between the reacting thermite (which can reach thousand Kelvin) and the surrounding water, localized boiling at the solid–liquid interface is likely to occur on very short timescales. An order-of-magnitude comparison suggests that the volume of vapor generated by water phase change can be comparable to, or even exceed, that produced solely from PDMS decomposition under the present conditions. Therefore, water vapor is likely a major component of the bubble, particularly during the early stages of expansion.

However, due to rapid cooling and condensation processes indicated by the simulations, the relative contribution of water vapor is expected to decrease during later stages of bubble evolution. In the absence of direct gas composition measurements, the bubble is interpreted as a mixture of water vapor and reaction-generated gases, including PDMS decomposition products and possible thermite-related species. While the presence of water vapor is strongly supported by thermal considerations, the precise composition and relative contributions of these gas species remain uncertain.

It should be emphasized that the proposed gas-generation pathways are qualitative, as the chemical composition of the bubble cannot be determined from the present data. A definitive understanding would require direct experimental characterization, such as gas chromatography, mass spectrometry, or in-situ spectroscopic diagnostics, which are beyond the scope of this study.

A different simulation by Bellama et al. propose that smaller molecules could be generated under rapid heating conditions³², and PDMS decomposition in infrared irradiation might produce methane or ethane through methyl radicals generated by Si-C bond cleavage^{32,33}. Considering their measured methane-ethane ratio and assuming complete methyl detachment, in the most ideal case, the maximum volume of gas from PDMS decomposition would be approximately 0.1 ml, which merely matches the observed gas volume, these calculations suggest that additional gas sources may contribute; however, direct measurements are required to confirm this hypothesis.

Bubble evolution mechanism

Based on the energy properties of the reagents and the simulations, a more detailed analysis on the evolution of the bubble could be made as follows:

Before the bubble appeared at 2.3 ms, the sample remained nearly motionless, undergoing continuous heating with only negligible movements observed in the air pocket.

While the bubble that emerged at 2.3 ms could be water vapor, observations by Akhatov et al. revealed that a millimeter-sized bubble induced in water by a laser pulse collapses within a microsecond, and experiments by Kim et al. suggests that it takes seconds for focused continuous laser to induce a bubble in water³⁶. Considering the rapid heat transfer at the liquid/vapor interface summarized by Kim³⁷, it is suggested that the bubble comprised other gases. Considering that PDMS can decompose at approximately 170 °C³⁵, CuO starts to decompose at above 700–800 °C³⁸, and our DSG results demonstrated weight loss in the Al/CuO-PDMS composite starting at about 300 °C, the initial slow-expansion stage may be attributed to PDMS decomposition at this temperature. This endothermic decomposition was sustained by the external laser, and with the gas generated in this process, heat loss to the water was greatly reduced.

When the sample reached approximately 550 °C in Fig. 4d, the nanothermite reaction started. This exothermic reaction became self-sustaining, resulting in the rapid expansion of the bubble. Numerous events, including thermal expansion, water evaporation, release of dissolved gasses, PDMS decomposition into smaller molecules, and CuO decomposition into O₂, together pushing the bubble to its maximum size.

Despite the nanothermite reaction lasted until 28 ms as shown in Fig. 4n, o and completely ended in Fig. 4p, the average bubble size started to decrease 10 ms before the reaction ended. The bubble then underwent approximately 6 pulsations following this general trend, as discussed earlier. Furthermore, the bubble experiences destabilization, the spherical surface fractured into smaller pieces over time.

This pulsation is quite different from the typical dynamics of other underwater bubbles, which contract to a much smaller size before rebounding, and maintaining a spherical form in the first cycle^{39–42}. This may suggest that gas generation occurs throughout the entire reaction, distinguishing it from the free expansion of the bubble.

Analyzing the bubble evolution and the sources of gas generation clarifies the chemical reactions occurring in the millisecond-scale regime. These observations also point toward the need to investigate the potential electronic interactions that happen on much shorter timescales that may help initiate the chemical reactions.

Interfacial considerations and conceptual energy-band alignment

The optical measurements presented in the Supplementary Material indicate that the optical response of the Al/CuO nanothermite differs from that of its individual constituents. Specifically, changes in the measured optical spectra were observed after Al and CuO nanoparticles were combined to form the nanothermite system. These observations demonstrate that the nanothermite exhibits distinct optical behavior compared with the separate materials. However, the optical measurements alone do not provide direct evidence of electronic coupling, charge transfer, carrier transport, or other interfacial electronic processes.

CuO semiconductors can exhibit either p-type or n-type conductivity, as discussed by Aleksandar Živković and Nora H. de Leeuw⁴³. Previous investigations have predominantly reported p-type conduction in CuO, which is attributed to the presence of copper vacancies and interstitial oxygen in the solid^{44,45}. The work function of

p-type CuO is approximately 5.3 eV⁴⁶, whereas aluminum has a lower work function of about 4.1 eV⁴⁷. Based on these reported material properties, a conceptual equilibrium energy-band alignment at the Al/CuO interface can be constructed, as illustrated in Fig. 9^{48,49}.

Figure 9 is presented solely as a conceptual illustration based on reported material properties and literature work-function values. It is intended to provide a qualitative visualization of the Al/CuO interface and should not be interpreted as representing experimentally observed electronic processes, carrier dynamics, or interfacial energy-transfer mechanisms. Under laser irradiation, the nanothermite absorbs optical energy, as evidenced by the measured optical response. However, the present study does not determine how the absorbed energy is distributed at the nanoscale or whether specific interfacial electronic interactions occur. Consequently, no conclusions regarding electronic excitation, charge transfer, carrier transport, localized energy deposition, or reaction-initiation mechanisms can be drawn from the present measurements.

Compared with micron-scale thermites, nanothermites possess substantially larger specific interfacial areas and a higher density of Al/CuO contacts. The optical measurements demonstrate that the nanothermite exhibits a different optical response from its individual constituents. While these observations indicate that the presence of Al/CuO interfaces is associated with altered optical behavior, the present measurements do not establish the microscopic origin of these differences or identify the underlying physical mechanisms.

Therefore, the discussion associated with Fig. 9 should be regarded as a qualitative interpretation based on established material properties reported in the literature and the experimentally observed optical behavior of the Al/CuO nanothermite. It is not intended as evidence of specific electronic, carrier-mediated, or interfacial energy-transfer processes. Elucidation of the underlying microscopic mechanisms would require direct experimental characterization of electronic structure and carrier dynamics, which is beyond the scope of the present work.

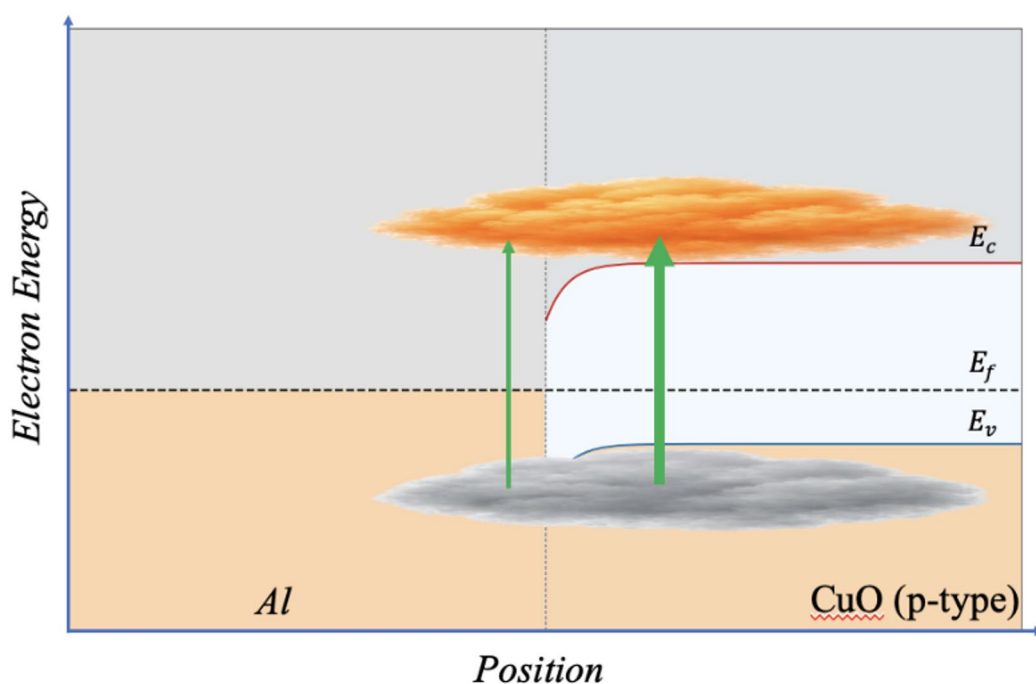


Fig. 9 Conceptual illustration of a possible equilibrium energy-band alignment at the Al/CuO (p-type) interface based on reported material properties. Occupied electronic states are shown in orange and unoccupied electronic states in gray. The band gap is represented in white, where E_c denotes the conduction-band minimum, E_v the valence-band maximum, and E_f the equilibrium Fermi level. Green arrows schematically represent possible laser-energy absorption near the nanoparticle contact region. This figure is provided only as a qualitative visualization of potential interfacial interactions and is not intended to represent experimentally observed electronic processes or carrier dynamics.

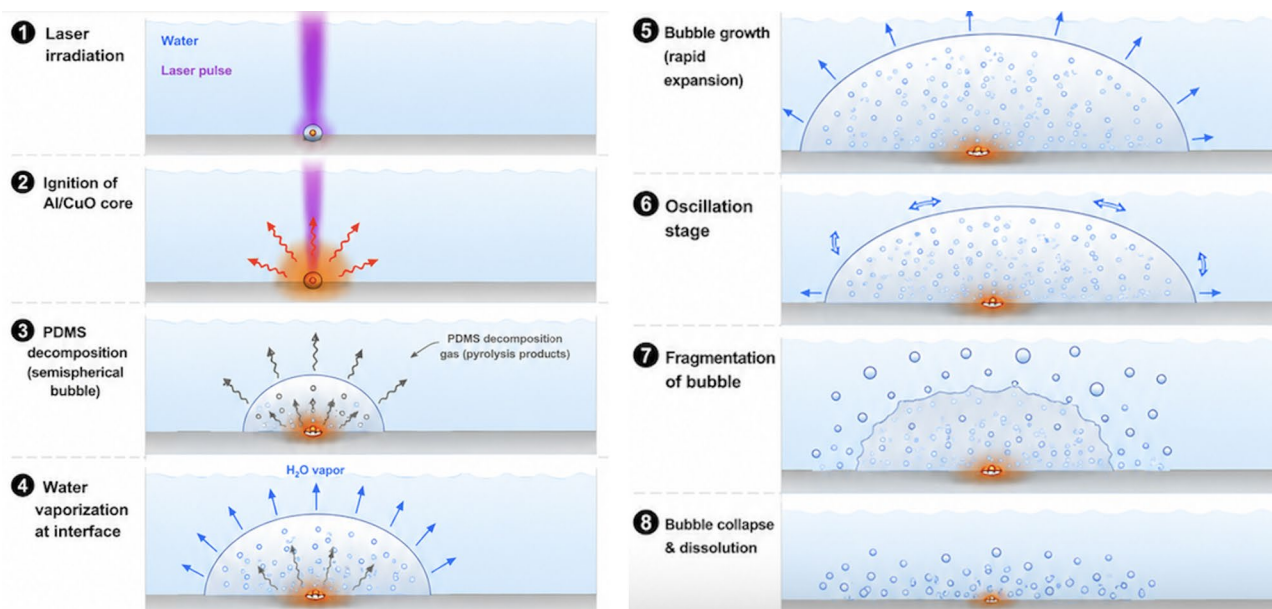


Fig. 10 Schematic illustration of the laser-triggered formation, evolution, and collapse of a bubble generated by an Al/CuO nanothermite embedded in PDMS at a water–solid interface. (1) Incident laser irradiation passes through the water and PDMS with negligible absorption and is primarily absorbed by the Al/CuO core. (2) Localized heating initiates ignition of the Al/CuO nanothermite reaction, releasing substantial thermal energy. (3) The surrounding PDMS undergoes rapid pyrolysis, producing gaseous decomposition products and forming an initial hemispherical gas bubble at the interface. (4) Heat transfer from the reaction zone to the adjacent water causes intense vaporization, and the generated water vapor mixes with the pyrolysis gases, further inflating the bubble. (5) Continued heat release and gas generation rapidly increase the internal pressure, driving dramatic bubble expansion into a large hemispherical cavity. (6) The bubble undergoes oscillatory motion due to the balance between internal pressure and the inertia of the surrounding liquid. (7) As the reaction weakens and cooling progresses, gas generation decreases and vapor condensation lowers the internal pressure, causing bubble instability and fragmentation into smaller daughter bubbles. (8) Continued heat dissipation and gas loss lead to bubble collapse, followed by the gradual dissolution of the remaining microbubbles into the surrounding liquid.

Proposed mechanism of laser-induced bubble evolution

Figure 10 summarizes the proposed bubble evolution mechanism derived from the experimental observations and the preceding discussion, highlighting the coupled effects of nanothermite heat release, PDMS decomposition, and interfacial water vaporization that govern bubble nucleation, growth, oscillation, fragmentation, and collapse. Upon laser irradiation, the incident energy is primarily absorbed by the embedded Al/CuO nanothermite particle, triggering a highly exothermic reaction that rapidly heats the surrounding PDMS matrix and initiates gas generation. The combined effects of PDMS pyrolysis and interfacial water vaporization lead to the formation and expansion of a hemispherical bubble, whose subsequent evolution is governed by the interplay among heat release, gas production, vapor condensation, and the inertia of the surrounding liquid.

The proposed mechanism consists of eight stages. First, the incident laser energy passes through the water and PDMS with negligible absorption and is predominantly absorbed by the Al/CuO nanothermite core. Second, localized heating initiates the thermite reaction, releasing a substantial amount of heat. Third, the surrounding PDMS undergoes rapid pyrolysis, generating gaseous decomposition products and forming a hemispherical gas dome at the water–PDMS interface. Fourth, heat transfer from the reaction zone to the adjacent water causes intense vaporization, and the generated water vapor mixes with the pyrolysis gases, resulting in further bubble inflation. Fifth, continued heat release and gas generation rapidly increase the internal pressure, driving dramatic hemispherical bubble expansion. Sixth, the bubble undergoes oscillatory motion due to the competition between internal pressure and the inertia of the surrounding liquid. Seventh, as the reaction subsides and heat dissipates,

gas generation decreases and vapor condensation lowers the internal pressure, causing the bubble to become unstable and fragment into smaller daughter bubbles. Finally, continued cooling and gas loss lead to bubble collapse and the gradual dissolution of the remaining microbubbles into the surrounding liquid.

Limitations and future work

Introducing PDMS and the surrounding water environment further increases the complexity of the system by adding additional thermal, chemical, and transport processes, including potential gas generation. Although unlikely to dominate under the present conditions, it is theoretically possible for aluminum to react with water vapor to form hydrogen⁵⁰, and such secondary pathways cannot be completely excluded.

Overall, a more definitive understanding of the underlying mechanisms would require additional experimental investigation, such as in-situ spectroscopic diagnostics, emission measurements, or direct analysis of reaction intermediates and products. Future work incorporating such techniques would help clarify the respective roles of interfacial processes, thermal effects, and gas generation in the ignition and combustion behavior.

The present study focuses on demonstrating underwater ignition feasibility and characterizing the associated bubble dynamics, which are markedly different from combustion behavior in air or vacuum. While these observations provide important insight into the ignition process, a comprehensive assessment of combustion performance—including ignition-energy thresholds, reaction duration statistics, propagation behavior, pressure generation, and energy efficiency—remains beyond the scope of the current work. Such quantitative characterization is an important direction for future research and will be pursued to provide a more complete understanding of underwater nanothermite combustion.

Summary

By dispersing PDMS prepolymers and Al/CuO nanothermite powders in a solvent such as acetone, a hydrophobic PDMS coating was formed around the nanothermite particles, enabling laser ignition underwater. The hydrophobic coating plays a critical role by reducing heat loss to the surrounding water and facilitating energy accumulation for ignition. The water contact angle increased from 115° to 150° as the PDMS content increased from 1% to 12%; however, only samples containing 2–9% PDMS could be successfully ignited underwater.

This study further investigated the heat-transfer and ignition mechanisms of the Al/CuO nanothermite system. Analysis based on the Rayleigh–Plesset equation, together with experimental observations and thermodynamic considerations, suggests rapid cooling and condensation of water vapor during bubble evolution, indicating that the bubble likely contains both water vapor and reaction-generated gases.

These findings improve the understanding of heat transport and ignition processes in nanothermite systems and demonstrate an effective strategy for achieving underwater ignition through hydrophobic surface modification. The coating approach may be extended to other energetic materials and hydrophobic polymers. In addition, the observed optical behavior and large interfacial area of the Al/CuO nanothermite suggest that interfacial effects may influence energy absorption and localization during laser ignition, although the detailed underlying mechanisms require further experimental investigation.

Supplementary Information The online version contains supplementary material available at <https://doi.org/10.1038/s41598-026-68857-0>.

Acknowledgements M.S. and Z.W. acknowledge the support of the UMass Lowell Internal Seed Awards Program.

Author contributions Author Contributions Z.W., J.W., S.D. and M.S. conceived the idea for the project. S.D. and S.K. provided equipment and participated the discussion. Z.W. demonstrated the feasibility, A.W. and J.E. improved the formula. Z.W., A.W. and J.E. performed the experiments. L.S. measured and analyzed the optical properties of the samples. Z.W.,

A.W., J.E., J.Z. and M.S. discussed the results, analyzed the data, and drafted the manuscript. J.W. and M.S. finalized the manuscript.

Funding This research was supported by UMass Lowell Internal Seed Awards. (Project ID: R50220000054421).

Data availability All data generated or analysed during this study are included in this published article [and its supplementary information file.

Declarations

Competing interests The authors declare no competing interests.

Open Access This article is licensed under a Creative Commons Attribution-NonCommercial-NoDerivatives 4.0 International License, which permits any non-commercial use, sharing, distribution and reproduction in any medium or format, as long as you give appropriate credit to the original author(s) and the source, provide a link to the Creative Commons licence, and indicate if you modified the licensed material. You do not have permission under this licence to share adapted material derived from this article or parts of it. The images or other third party material in this article are included in the article's Creative Commons licence, unless indicated otherwise in a credit line to the material. If material is not included in the article's Creative Commons licence and your intended use is not permitted by statutory regulation or exceeds the permitted use, you will need to obtain permission directly from the copyright holder. To view a copy of this licence, visit <http://creativecommons.org/licenses/by-nc-nd/4.0/>.

References

1. Saceleanu, F. et al. Combustion characteristics of physically mixed 40 nm aluminum/copper oxide nanothermites using laser ignition. *Front. Chem.* **6**, 465 (2018).
2. Li, H. et al. The effect of alumino-thermic addition on underwater wet welding process stability. *J. Mater. Process. Technol.* **245**, 149–156 (2017).
3. Wang, J. et al. The efficiency of thermite-assisted underwater wet flux-cored arc welding process: Electrical dependence, microstructural changes, and mechanical properties. *Metals* **13**(5), 831 (2023).
4. Strohmeier, B. R. Improving the wettability of aluminum foil with oxygen plasma treatments. *J. Adhes. Sci. Technol.* **6**(6), 703–718 (1992).
5. Nam, Y. & Ju, Y. S. Comparative study of copper oxidation schemes and their effects on surface wettability. In *ASME International Mechanical Engineering Congress & Exposition* (2008).
6. Valluri, S. K., Schoenitz, M. & Dreizin, E. Fluorine-containing oxidizers for metal fuels in energetic formulations. *Defence Technol.* **15**(1), 1–22 (2019).
7. Wu, T. et al. Performance of iodine oxides/iodic acids as oxidizers in thermite systems. *Combust. Flame* **191**, 335–342 (2018).
8. Knapp, S. et al. Emission spectroscopy of the combustion flame of aluminium/copper oxide thermite. *Propellants Explos. Pyrotech.* **44**(1), 9–17 (2019).
9. Fischer, S. H. & Grubelich, M. *Theoretical Energy Release of Thermites, Intermetallics, and Combustible Metals* (Sandia National Lab. (SNL-NM), 1998).
10. Sullivan, K. & Zachariah, M. Simultaneous pressure and optical measurements of nanoaluminum thermites: Investigating the reaction mechanism. *J. Propuls. Power* **26**(3), 467–472. <https://doi.org/10.2514/1.45834> (2010).
11. Williams, R. A. et al. Correlation of optical emission and pressure generated upon ignition of fully-dense nanocomposite thermite powders. *Combust. Flame* **160**(3), 734–741 (2013).
12. Weismiller, M. et al. Dependence of flame propagation on pressure and pressurizing gas for an Al/CuO nanoscale thermite. In *Proceedings of the Combustion Institute*. Vol. 32(2). 1895–1903 (2009).
13. Egorshv, V., Sinditskii, V. & Yartsev, K. Combustion of high-density CuO/Al nanothermites at elevated pressures. In *Autumn Seminar (International) on Propellants, Explosives and Pyrotechnics Proceedings* (2013).
14. Sullivan, K. et al. Reactive sintering: An important component in the combustion of nanocomposite thermites. *Combust. Flame*. **159** (1), 2–15 (2012). Retrieved from.
15. Meir, Y. & Jerby, E. Underwater microwave ignition of hydrophobic thermite powder enabled by the bubble-marble effect. *Appl. Phys. Lett.* **107**(5) (2015).
16. Shi, Z. et al. Dynamic contact angle hysteresis in liquid bridges. *Colloids Surf. A Physicochem. Eng. Asp.* **555**, 365–371 (2018).

17. Mata, A., Fleischman, A. J. & Roy, S. Characterization of polydimethylsiloxane (PDMS) properties for biomedical micro/nanosystems. *Biomed. Microdev.* **7**, 281–293 (2005).
18. Sabbah, A., Youssef, A. & Damman, P. Superhydrophobic surfaces created by elastic instability of PDMS. *Appl. Sci.* **6**(5), 152 (2016).
19. Li, S. et al. Drag-reductive and anti-corrosive superhydrophobic surface fabricated on aluminum with thin PDMS/SiO₂ coating. *Mater. Res. Exp.* **6**(10), 1065a8 (2019).
20. Butt, H. J. et al. Contact angle hysteresis. *Curr. Opin. Colloid Interface Sci.* **59**, 101574. <https://doi.org/10.1016/j.coci.2022.101574>. <https://www.sciencedirect.com/science/article/pii/S1359029422000139> (2022).
21. Samaha, M. A., Tafreshi, H. V. & Gad-el-Hak, M. Superhydrophobic surfaces: From the lotus leaf to the submarine. *C. R. Méch.* **340**(1–2), 18–34 (2012).
22. Spathi, C. et al. A simple method for preparing super-hydrophobic powder from paper sludge ash. *Mater. Lett.* **142**, 80–83 (2015).
23. Abbot, M. et al. Minimal addition of fumed nanoparticles suppresses droplet splashing. *Appl. Phys. Lett.* **127**(8). <https://doi.org/10.1063/5.0268107> (2025).
24. Abbot, M. et al. Nanoparticles do not influence droplet break-up, spreading, or splashing. *J. Colloid Interface Sci.* **693**, 137570. <https://doi.org/10.1016/j.jcis.2025.137570>. <https://www.sciencedirect.com/science/article/pii/S0021979725009610> (2025).
25. Lauterborn, W. & Kurz, T. Physics of bubble oscillations. *Rep. Prog. Phys.* **73**(10), 106501. <https://doi.org/10.1088/0034-4885/73/10/106501> (2010).
26. Prosperetti, A. Vapor bubbles. *Annu. Rev. Fluid Mech.* **49**, 221–248. <https://doi.org/10.1146/annurev-fluid-010816-060221>. <https://www.annualreviews.org/content/journals/> (2017).
27. Minnaert, M. XVI. On musical air-bubbles and the sounds of running water. *Lond. Edinb. Dublin Philos. Mag. J. Sci.* **16**(104), 235–248 (1933).
28. Leighton, T. *The Acoustic Bubble*. 234–243 (Academic, 1994).
29. Camino, G., Lomakin, S. & Lazzari, M. Polydimethylsiloxane thermal degradation Part 1. Kinetic aspects. *Polymer* **42**(6), 2395–2402 (2001).
30. Camino, G., Lomakin, S. & Lageard, M. Thermal polydimethylsiloxane degradation. Part 2. The degradation mechanisms. *Polymer* **43**(7), 2011–2015 (2002).
31. Wen, X. et al. RTV silicone rubber degradation induced by temperature cycling. *Energies* **10**(7), 1054 (2017).
32. Bellama, J. M. & Manders, W. F. On the thermal decomposition of polydimethylsiloxanes. *Sci. Tot. Environ.* **73**(1–2), 87–93 (1988).
33. Manders, W. & Bellama, J. Multiphoton infrared laser-induced degradation of polydimethylsiloxane and hexamethyldisiloxane. *J. Polym. Sci. Polym. Chem. Ed.* **23**(2), 351–357 (1985).
34. Jian, G. et al. Nanothermite reactions: Is gas phase oxygen generation from the oxygen carrier an essential prerequisite to ignition? *Combust. Flame* **160**(2), 432–437 (2013).
35. Genovese, A. & Shanks, R. A. Fire performance of poly (dimethyl siloxane) composites evaluated by cone calorimetry. *Compos. Part A Appl. Sci. Manuf.* **39**(2), 398–405 (2008).
36. Kim, N., Park, H. & Do, H. Evolution of cavitation bubble in tap water by continuous-wave laser focused on a metallic surface. *Langmuir* **35**(9), 3308–3318 (2019).
37. Kim, J. Review of nucleate pool boiling bubble heat transfer mechanisms. *Int. J. Multiph. Flow.* **35** (12), 1067–1076 (2009).
38. Lupu, A. Thermogravimetry of copper and copper oxides (Cu₂O–CuO). *J. Therm. Anal. Calorim.* **2**(4), 445–458 (1970).
39. Byun, K. T., Kwak, H. Y. & Karng, S. W. Bubble evolution and radiation mechanism for laser-induced collapsing bubble in water. *Jpn. J. Appl. Phys.* **43**(9R), 6364 (2004).
40. Wang, J. et al. Bubble evolution mechanism of pulse spark discharge in seawater. *J. Phys. D Appl. Phys.* **52**(28), 285201 (2019).
41. Hsu, C. Y. et al. A numerical study on the underwater explosion bubble pulsation and the collapse process. *Ocean Eng.* **81**, 29–38 (2014).
42. Yingyu, C. et al. Study on the quantitative law of wall pressure caused by mini-charge underwater explosion bubble. *Ocean Eng.* **213**, 107552 (2020).
43. Živković, A. & de Leeuw, N. H. Exploring the formation of intrinsic p-type and n-type defects in CuO. *Phys. Rev. Mater.* **4**(7), 074606 (2020).
44. Chatterjee, S., Saha, S. K. & Pal, A. J. Formation of all-oxide solar cells in atmospheric condition based on Cu₂O thin-films grown through SILAR technique. *Solar Energy Mater. Solar Cells* **147**, 17–26 (2016).
45. Pavan, M. et al. TiO₂/Cu₂O all-oxide heterojunction solar cells produced by spray pyrolysis. *Solar Energy Mater. Solar Cells* **132**, 549–556 (2015).
46. Koffyberg, F. & Benko, F. A photoelectrochemical determination of the position of the conduction and valence band edges of p-type CuO. *J. Appl. Phys.* **53**(2), 1173–1177 (1982).
47. Lovas, F. & Snyder, L. *CRC Handbook of Chemistry and Physics*. (Internet Version 2008). Vol. 14(6) (CRC, 2008).

48. Pierret, R. F. *Semiconductor Device Fundamentals* (Pearson Education India, 1996).
49. Cui, S. *Metal-Semiconductors Contacts*. Online Resource Ed.: LibreTexts Engineering.
50. Puszynski, J. A. *Reactivity of Nanosize Aluminum with Metal Oxides and Water Vapor*. Vol. 800. AA6 (MRS Online Proceedings Library (OPL), 2003).

Publisher's note Springer Nature remains neutral with regard to jurisdictional claims in published maps and institutional affiliations.

Authors and Affiliations

Z. B. Wang¹ · A. Q. Wang² · J. Ehling²  · C. J. MacRobbie² · S. Kim³ · L. Saini¹ · S. Deng³  · J. Z. Wen²  · M. Y. Shen¹

✉ J. Z. Wen
john.wen@uwaterloo.ca

✉ M. Y. Shen
Mengyan_shen@uml.edu

¹ Department of Physics and Applied Physics, University of Massachusetts Lowell, 220 Pawtucket St, Lowell, MA 01854, USA

² Department of Mechanical and Mechatronics Engineering, University of Waterloo, 200 University Ave West, Waterloo, ON N2L 3G1, Canada

³ Department of Mechanical Engineering, Massachusetts Institute of Technology, 77 Massachusetts Ave., Building 3-336, Cambridge, MA 02139, USA